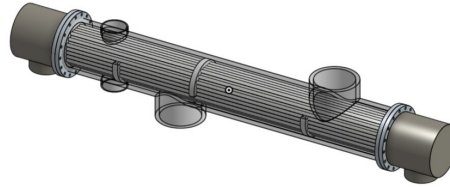


Jack Chobanian Engineering Portfolio 2026



I am a Chemical Engineering student at Cal Poly Pomona with a passion for propulsion, thermal systems, and mechanical design. I enjoy turning engineering concepts into physical designs by combining analytical problem solving with hands on experience. Outside the classroom, I have developed practical skills through automotive maintenance and repair while independently designing CAD projects such as a shell-and-tube heat exchanger. My goal is to continue developing as an engineer by creating designs that are functional, manufacturable, and grounded in real-world engineering principles. This portfolio showcases the projects that reflect my curiosity, technical growth, and commitment to becoming a chemical engineer.

831-710-7008 | jachobanian@cpp.edu | www.linkedin.com/in/jack-chobanian

Shell-and-Tube Heat Exchanger Design

Heat Exchanger | Onshape CAD

Objective: Designed a shell-and-tube heat exchanger using mechanical design practices.

Methodology

- Modeled the shell, tube bundle, tube sheets, baffles, channel heads, flanges, and nozzles.
- Designed a patterned tube bundle with equally spaced segmental baffles to improve shell-side flow.
- Assembled and constrained all components to create a realistic industrial assembly.

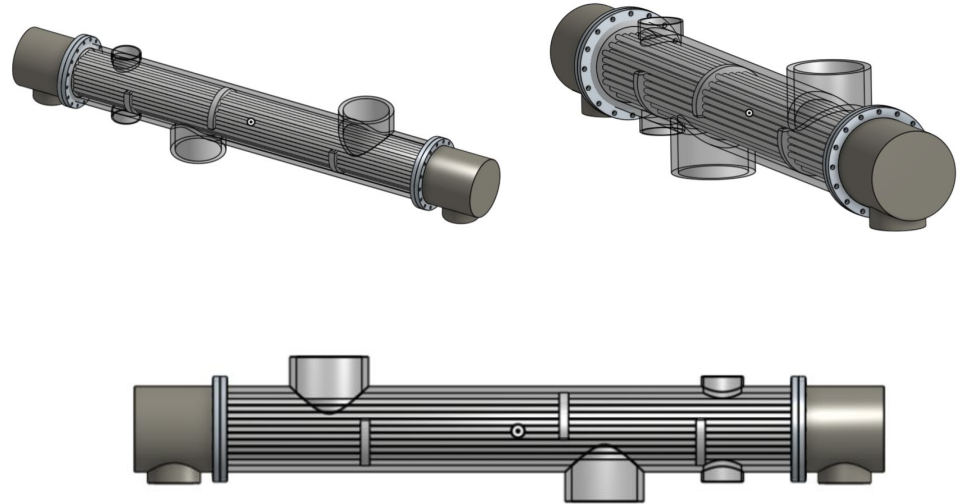
Results & Design Impact

- Completed a full industrial-style shell-and-tube heat exchanger assembly
- Gained understanding of heat exchanger principles

Key Skills

Software: Onshape CAD

Core Competencies: Mechanical Design, CAD Assembly,



Overall exchanger length: **144 in**

Shell length: **120 in**

Shell outside diameter: **24 in**

Shell wall thickness: **0.375 in**

Tube length: **116 in**

Tube outside diameter: **0.75 in**

Number of tubes: **48**

Baffle spacing: **24 in**

Baffle thickness: **0.25 in**

Shell-side nozzle diameter: **8 in**

Tube-side nozzle diameter: **6 in**

Flange outside diameter: **32 in**

(Conceptual dimensions selected for CAD demonstration; design is not intended for fabrication.)

